

PRE-BOARD EXAMINATION (2024-25)

CLASS X

MARKING SCHEME

Subject-Science (086)

1	D	1
2	C	1
3	A	1
4	D	1
5	B	1
6	C	1
7	B	1
8	C	1
9	C	1
10	D	1
11	C	1
12	A	1
13	A	1
14	C	1
15	D	1
16	D	1
17	D	1
18	A	1
19	C	1
20	B	1
21	Because bile is alkaline in nature and contains salts which helps to digest or emulsify fats	2
22	<u>Students to attempt either option A or B</u> A. Any food chain with at least three trophic level. If 2nd trophic level is removed –(i) three trophic level will starve and die, (ii) population of green plants will increase too much. OR B. It shields/ protects the surface of the earth from the entry of UV rays from the sun which can harm us by causing skin burnings and skin cancer.	1 ½+½ 1+1
23	<u>Students to attempt either option A or B</u> A. Esterification reaction $ \begin{array}{ccccccc} \text{C}_2\text{H}_5\text{OH} & + & \text{CH}_3\text{COOH} & \xrightarrow[\text{H}_2\text{SO}_4]{\text{Conc.}} & \text{CH}_3\text{COOC}_2\text{H}_5 & + & \text{H}_2\text{O} \\ \text{Ethanol} & & \text{Acetic acid} & & \text{Ethyl ethanoate} & & \\ \text{Or} & & \text{Or} & & \text{Or} & & \\ \text{Ethyl alcohol} & & \text{Ethanoic acid} & & \text{Ethyl acetate} & & \end{array} $ Ester formation (ethyl ethanoate) OR B. Surbhi's answer is correct as in saturated hydrocarbon heteroatom replaces hydrogen atom while in unsaturated hydrocarbon addition reaction occurs. Simple molecules are added across double or triple bond	½ 1 ½ 2
24	<u>Students to attempt either option A or B</u> A.(i) If connected in series total resistance will increase, current flowing through circuit will decrease (ii) If one appliance is switched off/ gets damaged all other appliances will stop working. OR B. - A.C-house, D.C -battery - A.C- long distance without much loss of heat	2 ½+½ ½+½

25	Shows tropic movement (directional response/ related to growth/ response of shoot towards light) while touch-me-not response is non-directional (nastic movement/ not related to growth/ due to loss of water by cells of the leaves)	1+1
26	It saves energy, reduces waste and pollution, conserves resources (any two)	2
27	<u>Students to attempt either option A or B</u> A.(i) When barium hydroxide is added into ammonium chloride, the bottom of test tube is found to be cooler. (ii) It is an endothermic reaction. (iii) $\text{Ba(OH)}_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{BaCl}_2 + 2\text{NH}_4\text{OH}$ OR B. i) a) Hydrogen gas released at the cathode b) Oxygen gas is released at anode ii) Because the water molecule contains two hydrogens and one oxygen in its molecular formula. Therefore, volume of one gas collected at one electrode is double (hydrogen gas) of the gas collected at the anode (oxygen gas). iii) Hydrogen gas - it burns with a pop sound.	1+1+1 $\frac{1}{2} + \frac{1}{2}$ 1 1
28	<u>Students to attempt either option A or B</u> A.(i) By increasing or decreasing the amount of water in them. When water enters the guard cells, they become turgid. This results in opening of stomatal pore. When water exits from the guard cells, they become flaccid. (ii) Vaseline will block the stomata and plant will become unhealthy due to the following reason. (any one reason) (a) There will be no gaseous exchange. (b) Vaseline layer hinders from receiving sunlight and affects the process of photosynthesis. No transpiration will take place and the plant will die. OR B.(i) Thin and large surface for easy diffusion, moist air sacs with mucus for dissolving the gases. The air sacs are surrounded by a network of blood capillaries with a good supply of blood so that more exchange of gases can happen. (ii) a) Diaphragm, b) Alveoli	1 1 1 1+1 $\frac{1}{2} + \frac{1}{2}$
29	(i) Mirror-1 concave; used by dentist/torch search light/ vehicle headlight Mirror-2 Convex; used as rear-view mirror / for security purpose (ii) Ratio of speed of light in air/vacuum to speed of light in the given medium, $n = c/v$	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ 1
30	(i) Calcium carbonate. CaCO_3 (ii) $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$	$\frac{1}{2} + \frac{1}{2}$ 1+1
31	<u>Students to attempt either option A or B</u> A.(i) MYOPIA (ii) $F = 1/p = 100/-4.5 = -22.2 \text{ cm or } -0.22 \text{ m}$ (iii) Concave lens OR B. (i) Statement made is incorrect. The position marked 3 (yellow) and 5 (blue) are similar to gold and colour of the sky. The students' statements are in reverse order. (ii) (a) position 7 (violet) (b) position 1 (red)	1+1+1 1+1 $\frac{1}{2} + \frac{1}{2}$
32	(i) Tt (ii) Law of dominance (iii) Out of 800 plants 600 plants will be tall and 200 plants will be short plants. Thus, in F₂ generation different traits are obtained in the ratio 3:1	1+1+1

33	A.(i) A solenoid is a coil of wire in the shape of cylinder with electric current flowing through it. Diagram (ii) Behave as electromagnet OR B. (i) So that any leakage of electric current is transferred to the ground or to prevent severe electric shock to the users. (ii) It is connected in series in domestic circuits. It prevents over heating of the wires and appliances due to excess transfer of electric current and melts when more current passes through it.	1+1+1 1 1+1
34	A.(i) $1/R_p = 1/R_1 + 1/R_2$ $1/5 + 1/10 = 3/10$ $\Rightarrow R_p = 10/3 \Omega$ (ii) $R_s = R_1 + R_2 + R_3$ $10 + 5 + 10 = 25 \Omega$ (iii) $1/R_p = 1/R_3 + 1/R_2$ $1/10 + 1/10 = 2/10$ $\Rightarrow R_p = 10/2 = 5 \Omega$ $R_{TOTAL} \Rightarrow 5 \Omega + 5 \Omega = 10 \Omega$ (iv) The potential difference V, across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same. This is called Ohm's law OR B.(i) length of wire, area of cross section, temp, nature of material (any two) $R = \rho l/A$ S.I unit of $\rho = \Omega m$ (ii) $\rho = R.A/l$ $\rho = 40 \times 6.5 \times 10^{-8} / 1$ $= 260 \times 10^{-8} \Omega m$ $= 2.6 \times 10^{-6} \Omega m$	1 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 2 $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1
35	<u>Students to attempt either option A or B</u> A.(i) Regeneration is carried out by specialized cells. These cells proliferate and make large numbers of cells. From this mass of cells, different cells undergo changes to become various cell types and tissues. Ex-Planaria (ii) testis and ovaries function: produce gamete and hormone vas deferens and oviduct/ fallopian tube- function-responsible to carry gamete for site of fertilizations OR B.(i) Two reasons- to avoid unwanted /undesirable pregnancies; prevent transmission of STDs (ii) changes hormonal balance of the body so that the eggs are not released. (iii) done for female foeticide. It adversely affects the male-female ratio	$\frac{1}{2} + 1\frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2}$ 2 1 1+1
36	<u>Students to attempt either option A or B</u> A.(i) Test tube-B (ii) Cu is lower to Iron in the activity series so displacement reaction will be maximum. (iii) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$	1+1+1

	(i) $\text{Mg} \longrightarrow \text{Mg}^{2+} + 2e^-$ 2, 8, 2 2, 8 (Magnesium cation) $\text{Cl} + e^- \longrightarrow \text{Cl}^-$ 2, 8, 7 2, 8, 8 (Chloride anion) $\text{Mg} \cdot + \cdot \text{Cl} \cdot \longrightarrow (\text{Mg}^{2+}) \left[\cdot \text{Cl} \cdot \right]_2^-$ OR B. (i) Any two metals- Zn/Fe; Displacement reaction of both metals as activity (ii) Calcination (iii) equations (both) $\text{ZnCO}_3 (\text{s}) \longrightarrow \text{ZnO} (\text{s}) + \text{CO}_2 (\text{g})$ $\text{ZnO} + \text{C} \longrightarrow \text{Zn} + \text{CO}$	1/2 1/2 1
	OR D. Receptor-sensory neuron-spinal cord (relay neuron)- motor neuron-muscles/gland	1+1/2+1 1 1+1
37	<u>Students to attempt either option C or D</u> A. Spinal cord B. axonal end/ / synaptic junction C. The electrical signal that travels down an axon is called a nerve impulse Slow, transmitted to only those body parts that are connected to neurons, in one direction only OR D. Receptor-sensory neuron-spinal cord (relay neuron)- motor neuron-muscles/gland	1 1 1 1/2+1/2 2
38	<u>Students to attempt either option C or D</u> A. Skeletal structure of ethanol $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ B. $\text{CO}_2 + \text{H}_2\text{O}$ C. $\text{CH}_3\text{CH}_2\text{OH} + [\text{O}] \xrightarrow[\text{KMnO}_4 + \text{heat}]{\text{Alkaline}} \text{CH}_3\text{COOH}$ acetic acid Role of alkaline potassium permanganate- oxidizing agent OR D. $\text{C}_2\text{H}_5\text{OH} \xrightarrow[443\text{K}]{\text{H}_2\text{SO}_4} \text{C}_2\text{H}_4 + \text{H}_2\text{O}$ ethene role of sulphuric acid- dehydration/ removal of water	1 1/2+1/2 1 1/2 1/2 1 1/2 1/2
39	<u>Students to attempt either option C or D</u> A. convex B. Magnifying glass, microscope (ANY ONE) C. OR D. Virtual, erect, magnified.	1+1+2